AttentionSurgeon: RL Agent Decides Which Attention Heads Live or Die

02501 Advanced Deep Learning in Computer Vision

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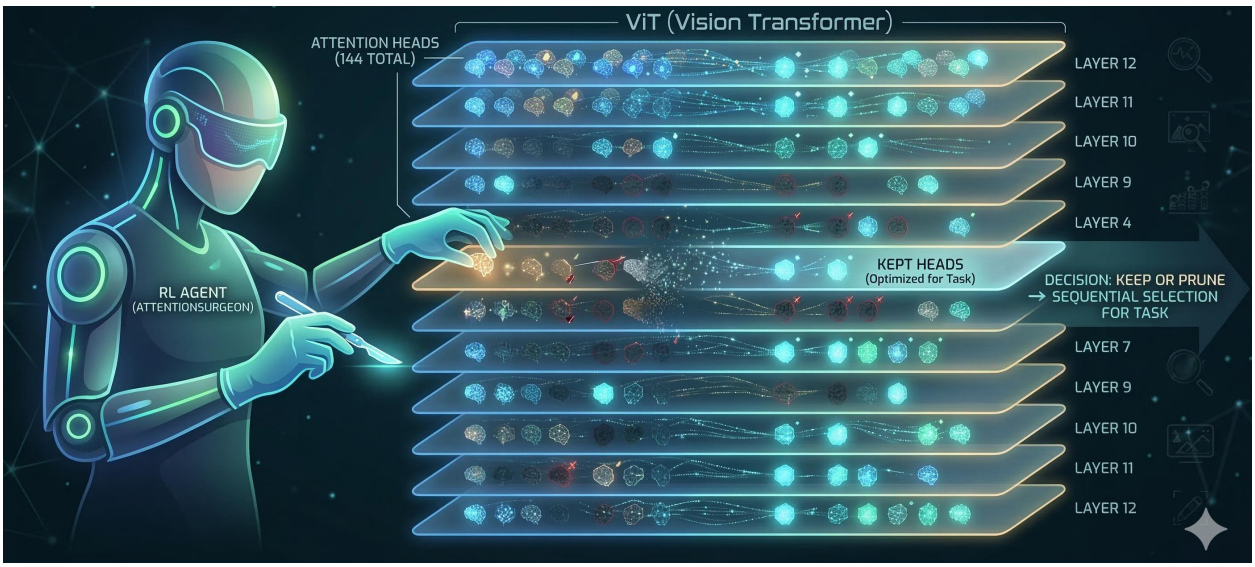

Figure 1: **AttentionSurgeon.**

1 Project description

A ViT-B/16 has 144 attention heads (12 layers $\times$ 12 heads). Recent mechanistic interpretability work [1, 2] has shown that many of these heads are redundant: some attend to CLS token globally, some track local edges, some are nearly dead. Pruning 30–50% of heads often barely affects accuracy [3]. But current pruning methods are task-agnostic: they remove the same heads regardless of whether the downstream task is classification, detection, or segmentation.

The idea of this project is to build AttentionSurgeon, an RL agent that learns which attention heads to keep for a specific downstream task, treating head selection as a sequential decision problem. The agent observes the model's current attention patterns and task performance, and decides which heads to prune next. The reward is: R = accuracy $\times$ FLOPs. This is an architecture

search via RL problem, but operating at an unusually fine granularity (individual attention heads) rather than the usual block-level NAS.

Different tasks need different heads. A classification model might keep mostly global/semantic heads and discard local ones, while a segmentation model needs the opposite (i.e., it must preserve heads that encode fine spatial relationships). No prior work has studied task-specific head selection via RL.

2 Data

Use a pre-trained DINOv2 ViT-B/16 [4] as the base model (144 attention heads, strong features for all tasks). Evaluate on three downstream tasks: (a) ImageNet-1K classification (linear probe), (b) ADE20K semantic segmentation [5] (linear probe on patch tokens), (c) COCO object detection (using the ViT as a feature extractor for a lightweight detection head). The RL agent is a small MLP policy network (state = 144-dim binary mask + task accuracy + FLOPs count; action = which head to prune next). Train the RL agent with PPO [6] or REINFORCE or GRPO. Each RL episode: start with all heads, sequentially prune one head at a time (up to 72), evaluate task accuracy after each pruning step, receive reward. The agent learns the optimal pruning order. All experiments should fit on 1 GPUs since we only train linear probes.

3 Tasks

In this project, you could work on the following tasks:

Task 1: Attention Head Census. Before RL, systematically characterize all 144 heads in DINOv2 ViT-B/16. For each head, compute: (a) average attention entropy (focused vs. diffuse), (b) average attention distance (local vs. global), (c) head importance via gradient-based attribution for each downstream task, (d) activation magnitude. Cluster heads into functional groups (e.g., local-edge, global-semantic, CLS-aggregator, positional, dead). Create a "periodic table of attention heads" visualization showing head properties across all 12 layers. Which heads are task-universal vs. task-specific?

Task 2RL-Guided Pruning Agent. Formulate the MDP: state = (binary mask of active heads, current task accuracy, current FLOPs), action = index of the next head to prune, reward = accuracy $\times$ FLOPs. Train separate agents for each task (classification, segmentation, detection). Compare the RL-discovered pruning order with: (a) random pruning, (b) magnitude-based pruning (remove lowest-activation heads first), (c) gradient-based pruning (remove least-important-by-gradient heads first), (d) uniform layer-wise pruning. Evaluate using accuracy vs. FLOPs Pareto curves for all methods.

Task 3: Cross-Task Head Analysis. Compare the head subsets selected by RL for each task. Compute the similarity between the "top-72 essential heads" for classification vs. segmentation vs. detection. Identify: (a) universal heads that all tasks keep (the "backbone" of the ViT), (b) classification-specific heads (global, semantic), (c) segmentation-specific heads (local, spatial), (d) detection-specific heads (multi-scale).

Task 4: ...

Task 5: ...

References

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